

GUATEMALA LA AURORA, Guatemala

WMO: 786410

Lat: 14.583N

Lon: 90.527W

Elev: 1509

StdP: 84.46

Time Zone: -6.00 (W06)

Period: 96-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	10.9	12.0	5.8	6.9	16.7	7.0	7.4	17.1	14.2	18.2	12.9	17.5	5.9	0	0.460

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
5	9.1	28.1	17.9	27.1	18.0	26.2	17.9	20.3	24.9	19.8	24.2	19.5	23.7	5.1	0

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
19.1	16.7	21.5	18.8	16.4	21.2	18.2	15.8	20.7	65.6	25.1	64.1	24.3	62.7	23.7	25.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
11.2	10.0	9.1	DB	8.8	30.3	1.1	1.0	8.0	31.0	7.4	31.6	6.8	32.2	6.0	32.9	(4)
			WB	6.9	22.0	1.2	1.2	6.1	22.9	5.3	23.6	4.7	24.2	3.8	25.1	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	20.2	18.3	19.4	20.3	21.6	21.7	20.9	20.7	20.9	20.7	20.1	18.9	18.6	(6)
		DBStd	1.67	1.60	1.49	1.43	1.21	1.16	1.03	0.90	0.83	1.01	1.24	1.57	1.48	(7)
		HDD10.0	0	0	0	0	0	0	0	0	0	0	0	0	0	(8)
		HDD18.3	58	20	7	2	0	0	0	0	0	1	2	12	14	(9)
		CDD10.0	3713	257	262	318	348	362	326	333	339	321	313	268	267	(10)
		CDD18.3	730	19	36	62	98	104	76	75	81	72	57	30	22	(11)
		CDH23.3	2259	77	158	307	474	381	153	166	196	132	92	53	71	(12)
(14)	Wind	WSAvg	4.9	6.1	5.7	5.3	4.7	4.0	3.8	5.0	4.4	3.4	4.5	6.0	6.0	(14)
		PrecAvg	1141	4	2	12	24	116	237	178	185	234	108	26	5	(15)
		PrecMax	1671	19	13	53	81	332	426	361	458	395	248	87	52	(16)
		PrecMin	813	0	0	0	0	45	107	51	37	98	12	0	0	(17)
		PrecStd	226	5	3	16	25	56	79	78	90	77	59	28	10	(18)
		DB	26.2	27.8	28.9	29.4	29.1	27.0	26.9	27.1	26.8	26.2	26.0	26.2	26.2	(19)
		MCWB	16.5	16.4	17.3	17.4	18.6	19.3	18.7	18.3	19.1	18.8	18.1	17.0	17.0	(20)
(22)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	2% DB	25.1	26.2	27.1	28.1	27.8	26.0	25.9	26.1	25.9	25.1	24.9	25.1	25.1	(21)
		2% MCWB	16.5	16.8	17.2	17.9	18.4	18.9	18.2	18.3	18.9	18.6	17.8	17.1	17.1	(22)
		5% DB	24.1	25.1	26.1	27.1	26.8	25.1	25.1	25.2	25.0	24.2	23.9	24.0	24.0	(23)
		5% MCWB	16.3	16.5	17.1	18.0	18.5	18.7	18.2	18.3	18.8	18.3	17.4	16.7	16.7	(24)
		10% DB	23.0	24.1	25.1	26.1	25.8	24.1	24.2	24.5	24.0	23.2	22.9	22.9	22.9	(25)
		10% MCWB	15.8	16.3	16.8	17.7	18.5	18.5	18.0	18.2	18.6	18.0	17.1	16.2	16.2	(26)
		WB	18.2	18.8	19.7	20.6	20.9	20.6	20.0	20.1	20.4	20.2	19.5	19.0	19.0	(27)
(28)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	MCDB	23.8	24.6	25.4	26.2	25.9	24.9	24.4	24.2	24.6	24.3	23.4	23.6	23.6	(28)
		2% WB	17.5	18.1	18.9	19.7	20.2	19.9	19.4	19.5	19.7	19.5	18.7	18.0	18.0	(29)
		2% MCDB	22.8	23.6	23.9	25.0	24.9	23.9	23.6	23.5	23.7	23.3	22.3	22.9	22.9	(30)
		5% WB	16.8	17.5	18.3	19.2	19.7	19.5	18.9	19.1	19.3	19.0	18.1	17.4	17.4	(31)
		5% MCDB	21.9	22.7	23.3	24.0	24.0	23.2	23.0	23.0	23.1	22.6	21.9	22.0	22.0	(32)
		10% WB	16.4	16.9	17.7	18.7	19.3	19.0	18.5	18.7	18.9	18.6	17.6	16.8	16.8	(33)
		10% MCDB	21.4	21.7	22.7	23.2	23.2	22.5	22.4	22.4	22.4	21.9	21.3	21.1	21.1	(34)
(35)	Mean Daily Temperature Range	MDBR	9.9	10.6	11.1	10.6	9.1	7.6	7.9	8.2	7.6	7.6	8.3	9.3	9.3	(35)
		5% DB	11.4	11.8	12.1	11.4	10.4	8.6	8.7	8.9	8.7	8.5	9.4	10.7	10.7	(36)
		5% MCWBR	4.4	4.3	4.4	3.9	3.8	3.3	3.1	3.3	3.3	3.4	3.9	4.3	4.3	(37)
		5% WB	10.5	11.2	11.0	10.6	9.4	8.0	8.0	8.2	8.0	7.7	8.2	9.8	9.8	(38)
		5% MCWBR	4.4	4.5	4.5	4.1	3.8	3.4	3.3	3.3	3.4	3.4	3.7	4.3	4.3	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.313	0.328	0.363	0.433	0.462	0.393	0.390	0.380	0.380	0.355	0.331	0.313	0.313	(40)
		taud	2.603	2.552	2.424	2.214	2.169	2.419	2.427	2.445	2.429	2.543	2.577	2.611	2.611	(41)
		Ebn at Noon	989	983	950	878	839	891	895	914	920	943	961	980	980	(42)
		Edh at Noon	94	103	120	148	152	117	117	116	118	103	96	91	91	(43)
(44)	All-Sky Solar Radiation	RadAvg	5.13	5.62	5.99	5.83	5.11	5.07	5.70	5.59	4.99	4.71	4.95	4.96	4.96	(44)
		RadStd	0.18	0.27	0.29	0.32	0.33	0.29	0.37	0.37	0.36	0.37	0.24	0.23	0.23	(45)

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (6 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page